Understanding and Modeling the Determinants of Residential Experience

Presented by Group W12_G03
Faculty of Engineering,
STAT5003 Computational Statistical Methods

THE UNIVERSITY OF SYDNEY

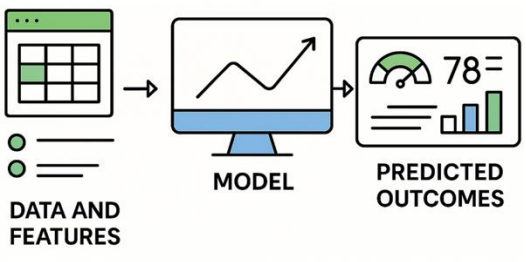
Motivation

Affordability
Convenient transportationand stable internet connectivity
Community safety and facilities suitable for children

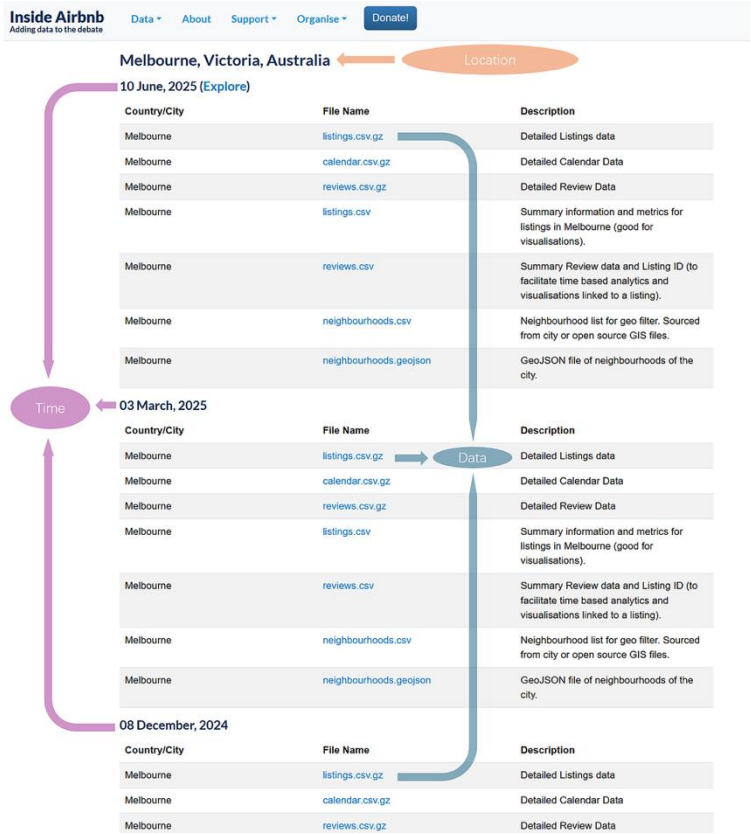
Research Question

Research Question One: Which factors significantly affect residential experience?

Research Question Two: How do different models perform in predicting the residential experience?



Data Acquisition



Data Preprocessing

1. Merge & Standardise

Merge: Automatically read and combined multiple CSV files split by city and date.

Clean: Removed irrelevant columns (e.g., IDs, URLs, descriptions).

Standardise:

- Unified host location name (e.g., "WA" -> "Perth").
- Cleaned non-numeric characters (e.g., "$" from price and "%" from host response rate variables).

Identify Target Variable: Discretised the continuous review scores into 5 review level categories (e.g., Very Poor, Poor, Moderate, Good, Excellent).

Data Preprocessing

2. Filter & Split

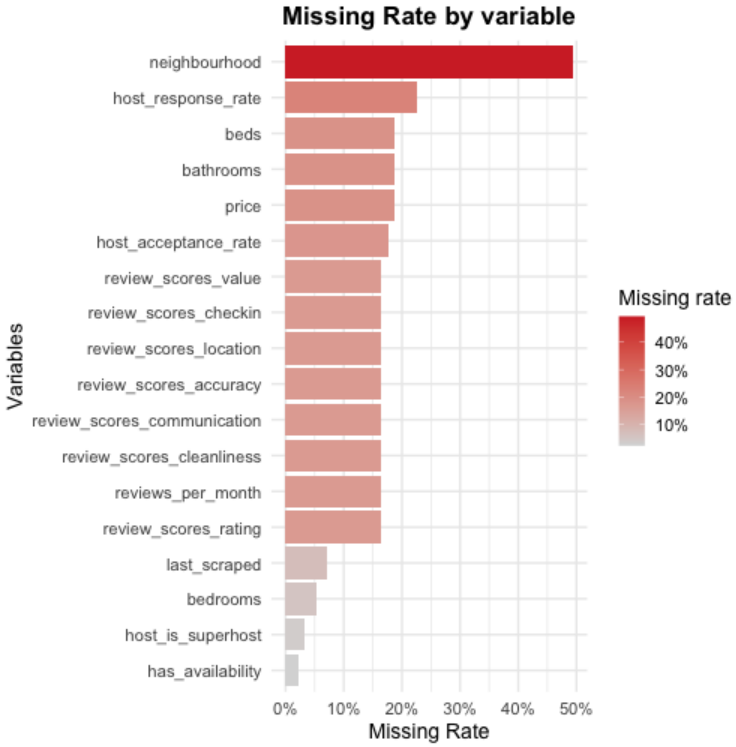
Filter Outliers: Removed "impossible" data based on business logic:

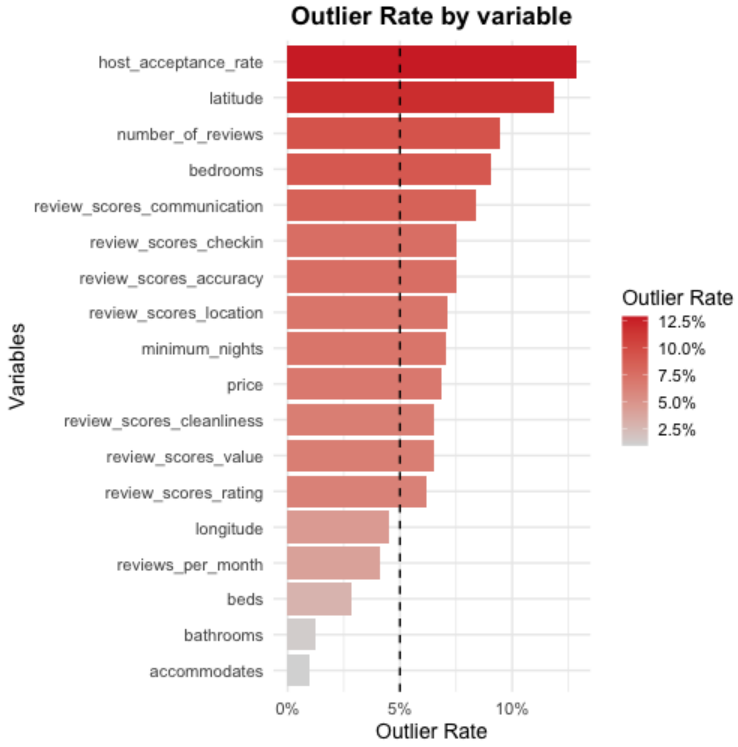
- *Invalid Values:* price < 0 or review_scores > 5.
- *Placeholders:* price = 99999.
- *Logical Errors:* Unreasonable accommodates to beds ratio.

Empty Rows Clean: Performed a strict method to remove all rows with missing values.

Model Preperation: Split the final dataset into 80% Training and 20% Testing sets.

Data Preprocessing





Data Description

Coverage:

- Adelaide, Brisbane, Melbourne, Sydney
- Tasmania (state-level), Western Australia (state-level)

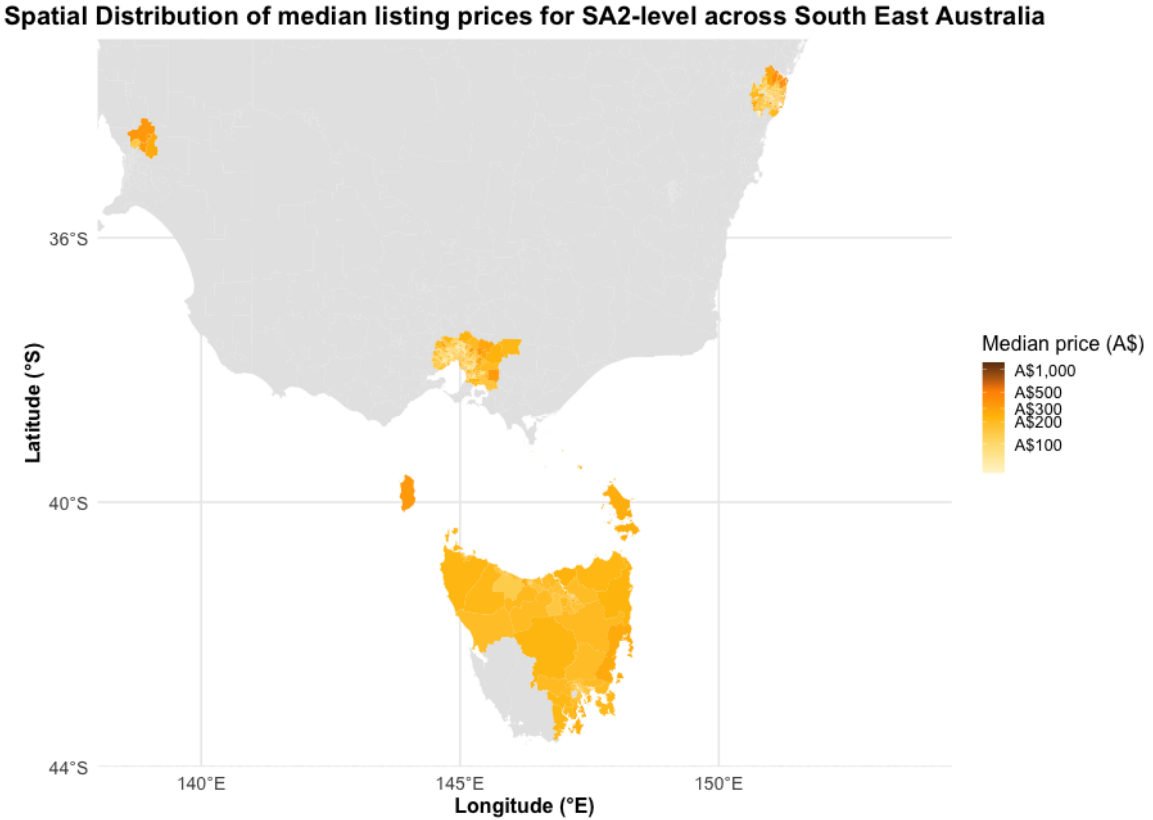
Time Periods: Sept 2024, Dec 2024, Mar 2025, Jun 2025

Variables: 75 original → 29 retained after data cleaning

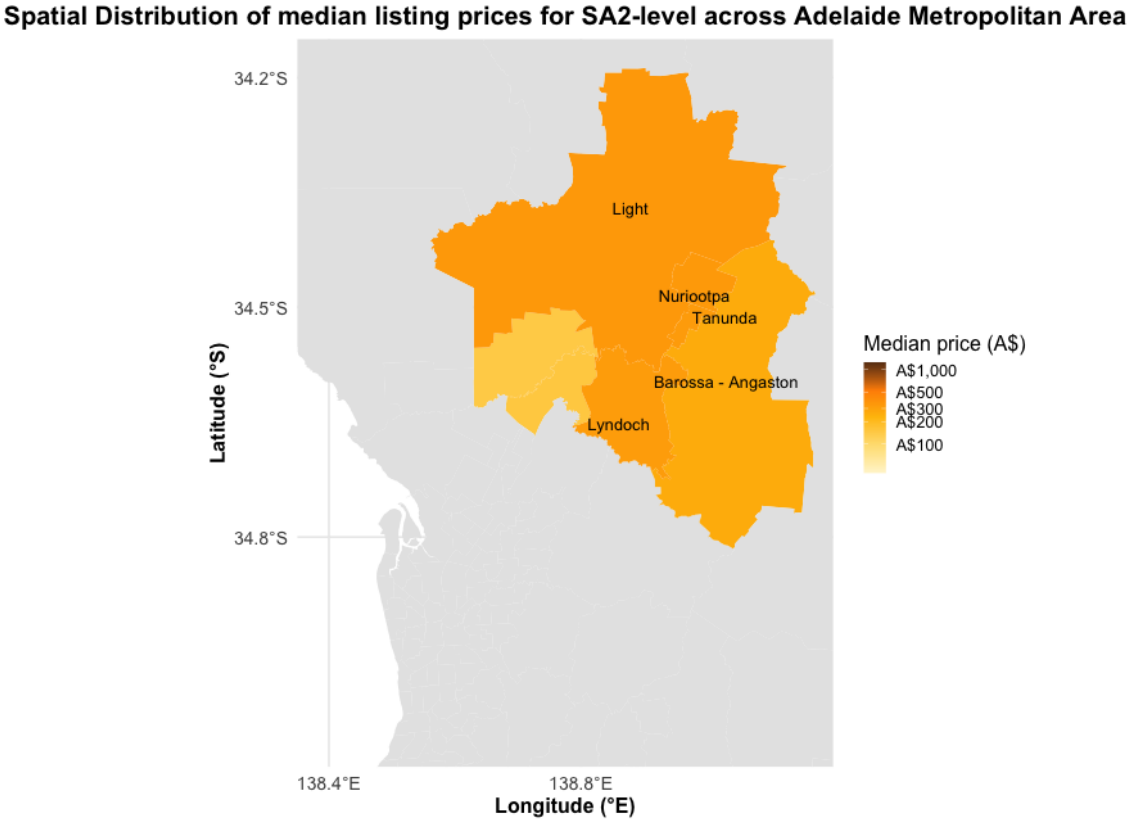
Records:

Stage	Records
Before cleaning	229,170
After cleaning	104,316

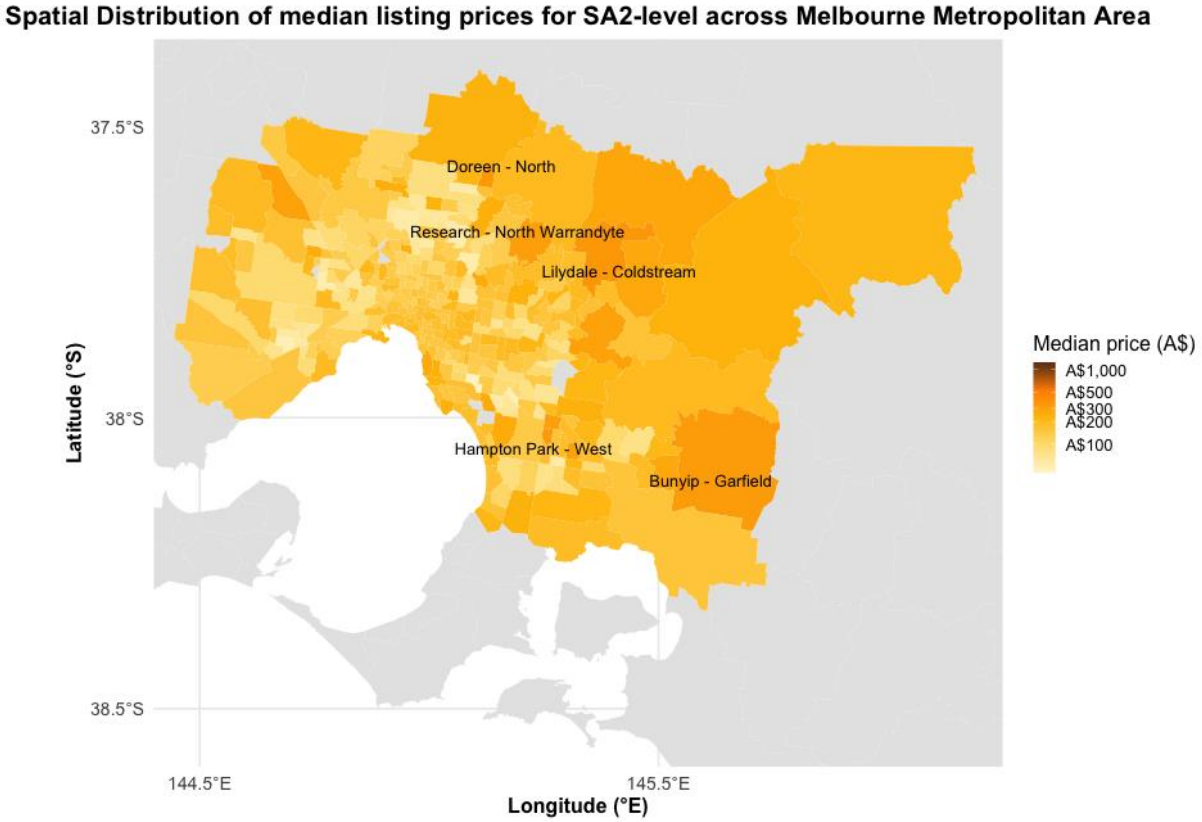
Data Visualisation



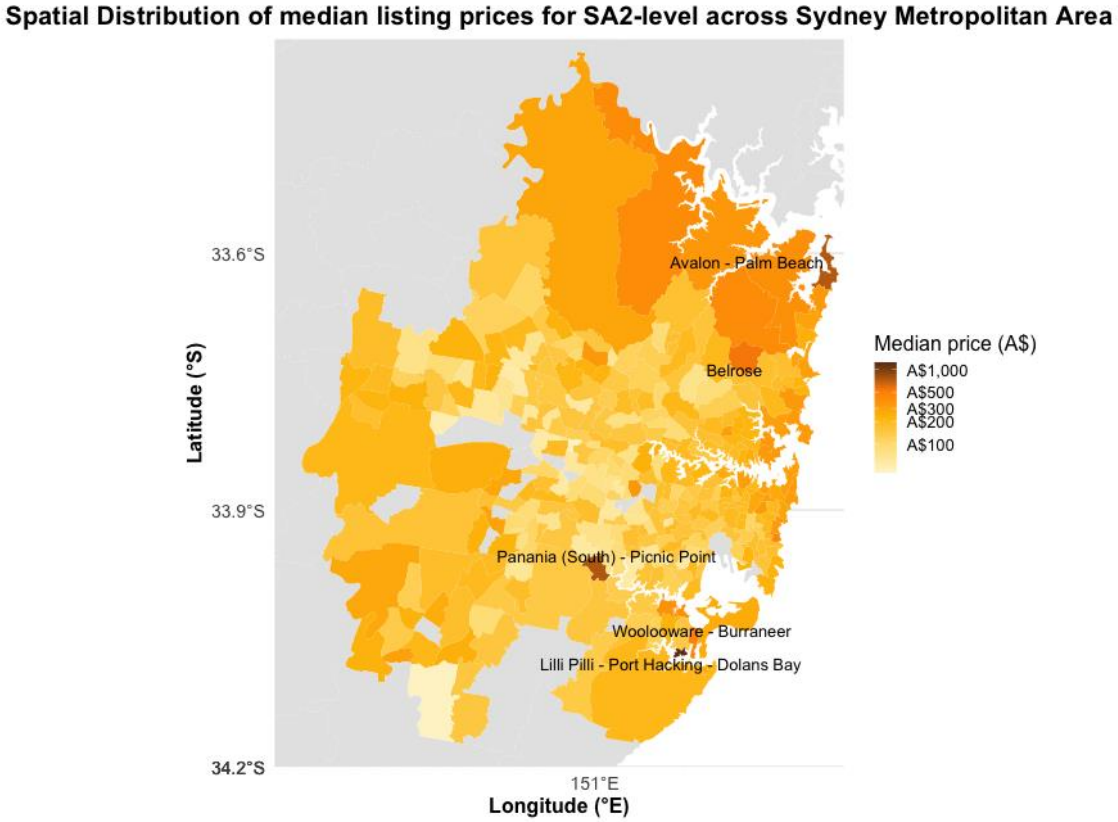
Data Visualisation



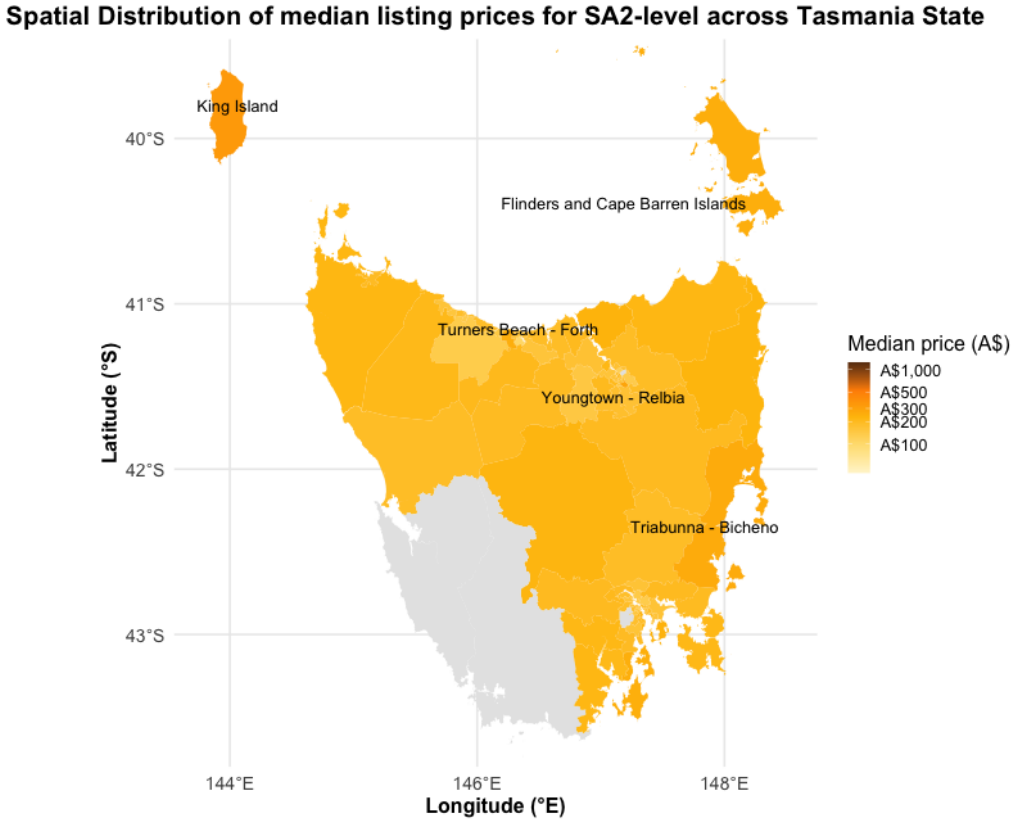
Data Visualisation



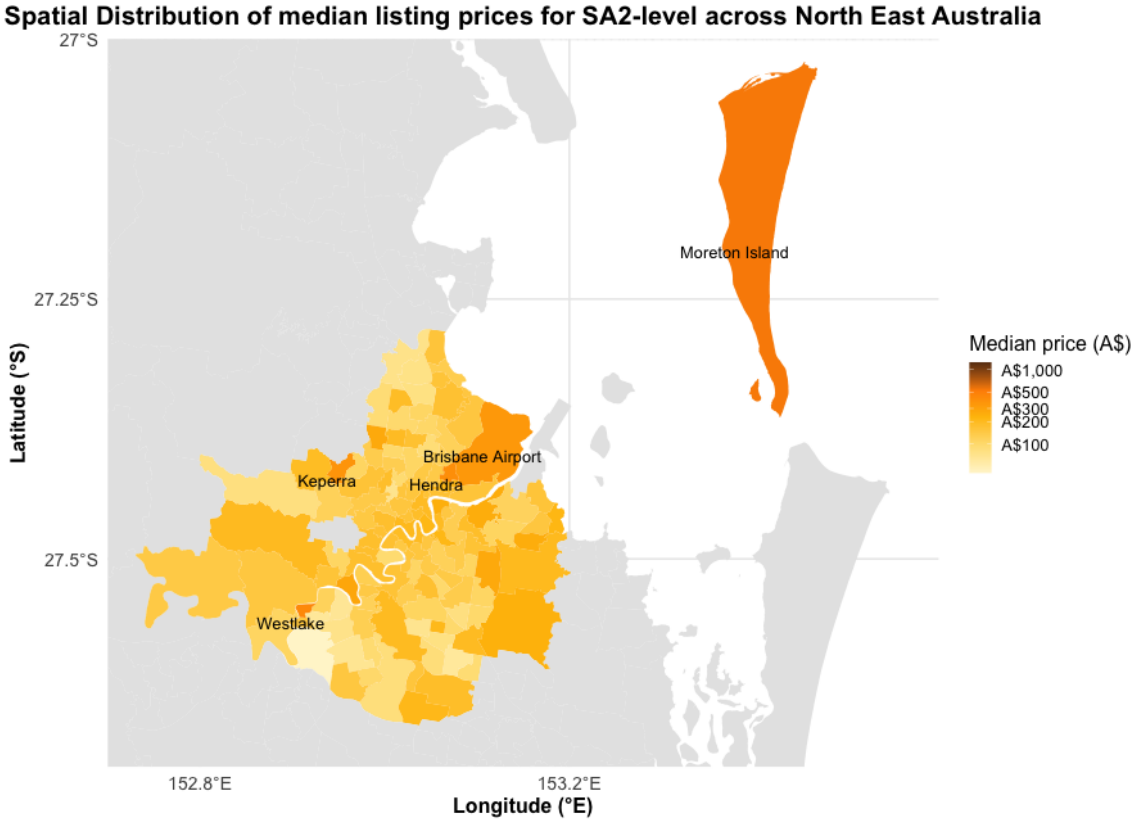
Data Visualisation



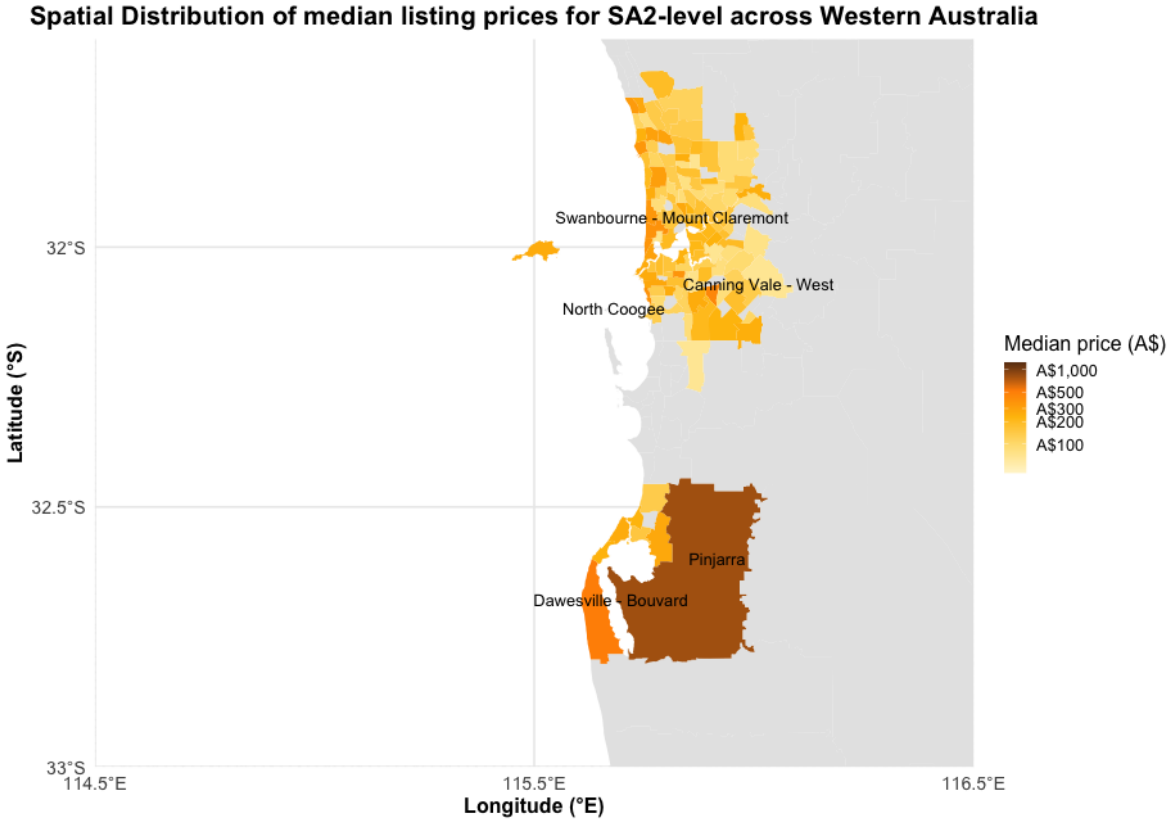
Data Visualisation



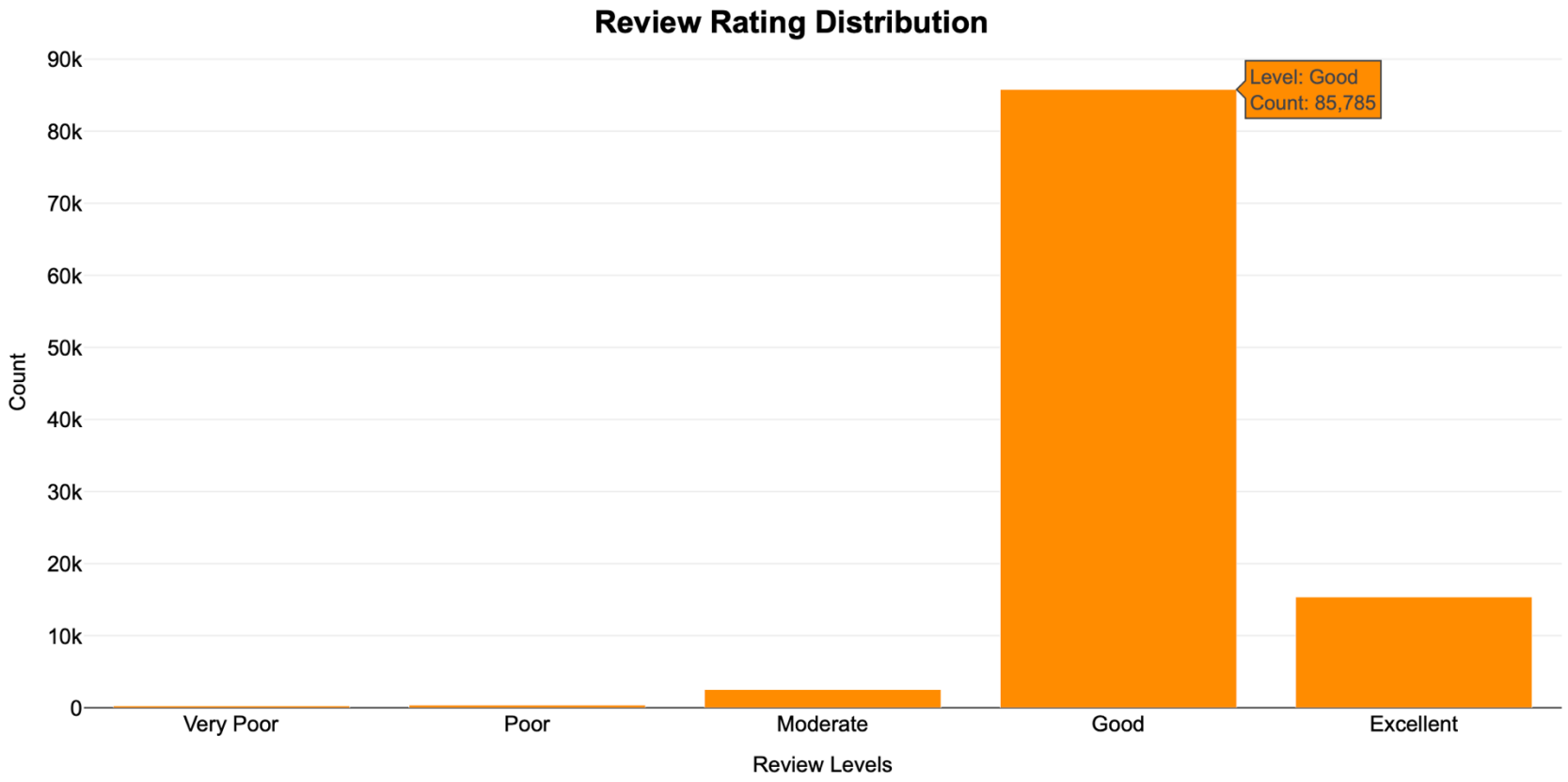
Data Visualisation



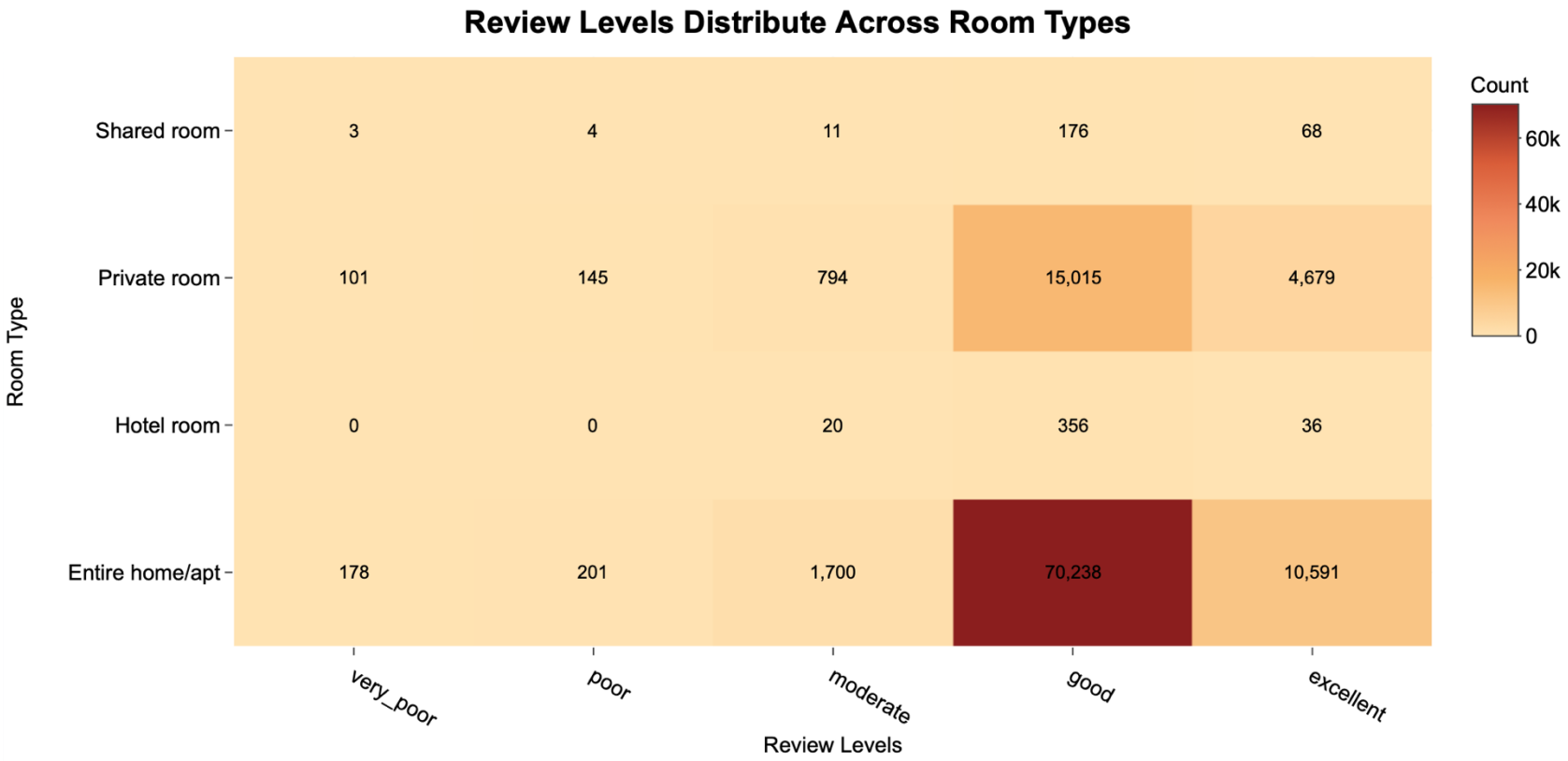
Data Visualisation



Data Visualisation



Data Visualisation



What Affects Residential Experience

Dependent variable: User rating with ordered categories: “Very Poor,” “Poor,” “Moderate,” “Good,” and “Excellent.”

Cumulative Link Model (CLM): The CLM is specifically designed for ordinal dependent variables.

Results of the Cumulative Link Model

Parameter	Estimate	Std_Error	z_value	p_value
Accomodates	0.002859	0.001768	1.617657	0.1058
House Location	-0.000664	0.000558	-1.190220	0.2341
House Has Availability	0.414603	1.033948	0.400990	0.6873
Bed Number	-0.000775	0.002495	-0.310689	0.7560

Predicting Residential Experience

Logistic Regression: Good interpretability, suitable for numerical features, naturally extends to multi-class (Softmax).

K-Nearest Neighbor (KNN): Non-parametric, good for medium-sized datasets, captures local differences based on distance.

Support Vector Machines (SVM): Effective with kernel tricks for non-linear boundaries, suitable for medium/small data.

Predicting Residential Experience

Linear Discriminant Analysis (LDA): Focuses on maximizing inter-class separation, good for dimensionality reduction and interpretability under Gaussian assumptions.

XGBoost: a powerful and efficient gradient boosting algorithm known for its high accuracy, speed, and ability to handle missing data and overfitting effectively.

Predicting Residential Experience

We are exploring a set of parameters for each models:

Model and Parameters		
Function	**Library**	**Parameters**
svmLinear3	LiblineaR	cost, Loss
lda2	MASS	dimen
xgbTree	xgboost	nrounds, max_depth, eta, gamma, colsample_bytree, min_child_weight, subsample
multinom	nnet	decay
knn	caret	k

Key Results and Visuals

	Linear Discriminant Analysis					Support Vector Machines				
	very_poor	poor	moderate	good	excellent	very_poor	poor	moderate	good	excellent
Recall	0.8036	0.6000	0.8376	0.8365	0.8936	0.9286	0.5143	0.5564	0.7067	0.8959
Precision	0.6923	0.4615	0.3490	0.9774	0.5711	0.5532	0.1682	0.1588	0.9748	0.4339
Specificity	0.9990	0.9976	0.9612	0.9104	0.8840	0.9980	0.9914	0.9269	0.9155	0.7980
Balanced Accuracy	0.9013	0.7988	0.8994	0.8735	0.8888	0.9633	0.7529	0.7417	0.8111	0.8470
Prevalence	0.0027	0.0034	0.0242	0.8224	0.1473	0.0027	0.0034	0.0242	0.8224	0.1473

Imbalance Challenge: High Prevalence of top ratings makes predicting rare low ratings difficult.

Minority Recall: Logistic Regression, Linear Discriminant Analysis & K-Nearest Neighbour show superior Recall for minority classes vs. XGBoost & Support Vector Machines.

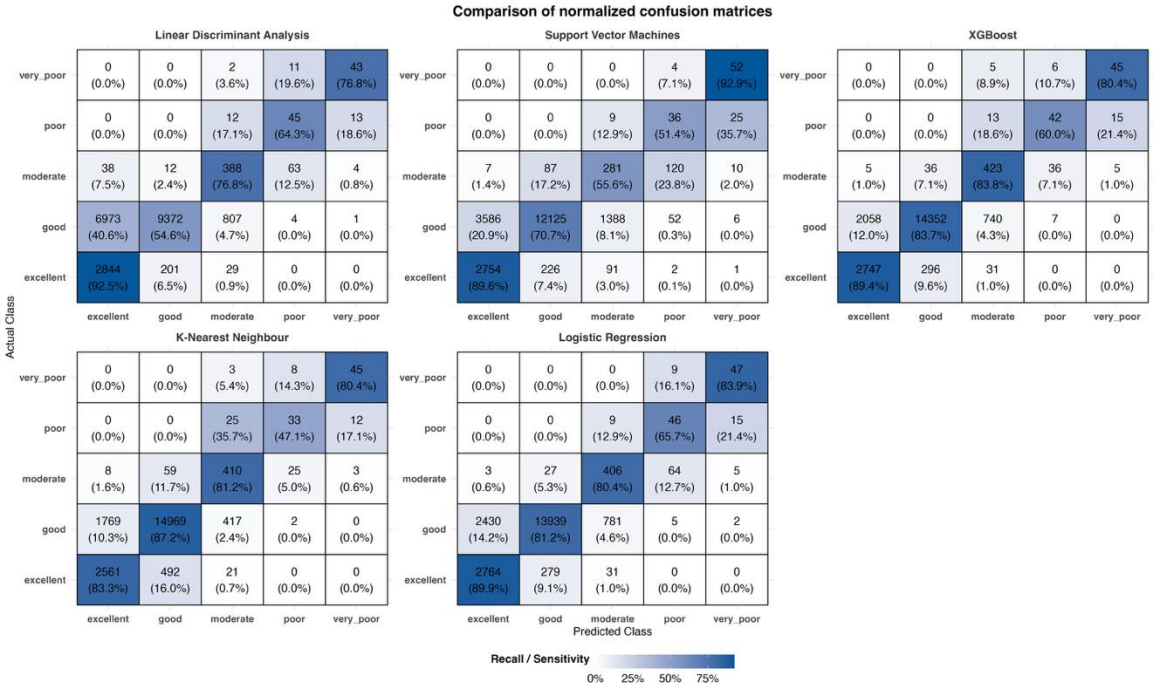
Key Results and Visuals

	XGBoost					K-Nearest Neighbour					Logistic Regression				
	very_poor	poor	moderate	good	excellent	very_poor	poor	moderate	good	excellent	very_poor	poor	moderate	good	excellent
Recall	0.7679	0.6429	0.7683	0.5462	0.9252	0.8036	0.4714	0.8119	0.8725	0.8331	0.8393	0.6571	0.8040	0.8124	0.8992
Precision	0.7049	0.3659	0.3134	0.9778	0.2886	0.7500	0.4853	0.4680	0.9645	0.5904	0.6812	0.3710	0.3309	0.9785	0.5318
Specificity	0.9991	0.9962	0.9583	0.9425	0.6059	0.9993	0.9983	0.9771	0.8513	0.9001	0.9989	0.9962	0.9597	0.9174	0.8632
Balanced Accuracy	0.8835	0.8196	0.8633	0.7444	0.7655	0.9014	0.7349	0.8945	0.8619	0.8666	0.9191	0.8267	0.8818	0.8649	0.8812
Prevalence	0.0027	0.0034	0.0242	0.8224	0.1473	0.0027	0.0034	0.0242	0.8224	0.1473	0.0027	0.0034	0.0242	0.8224	0.1473

Imbalance Challenge: High Prevalence of top ratings makes predicting rare low ratings difficult.

Minority Recall: Logistic Regression, Linear Discriminant Analysis & K-Nearest Neighbour show superior Recall for minority classes vs. XGBoost & Support Vector Machines.

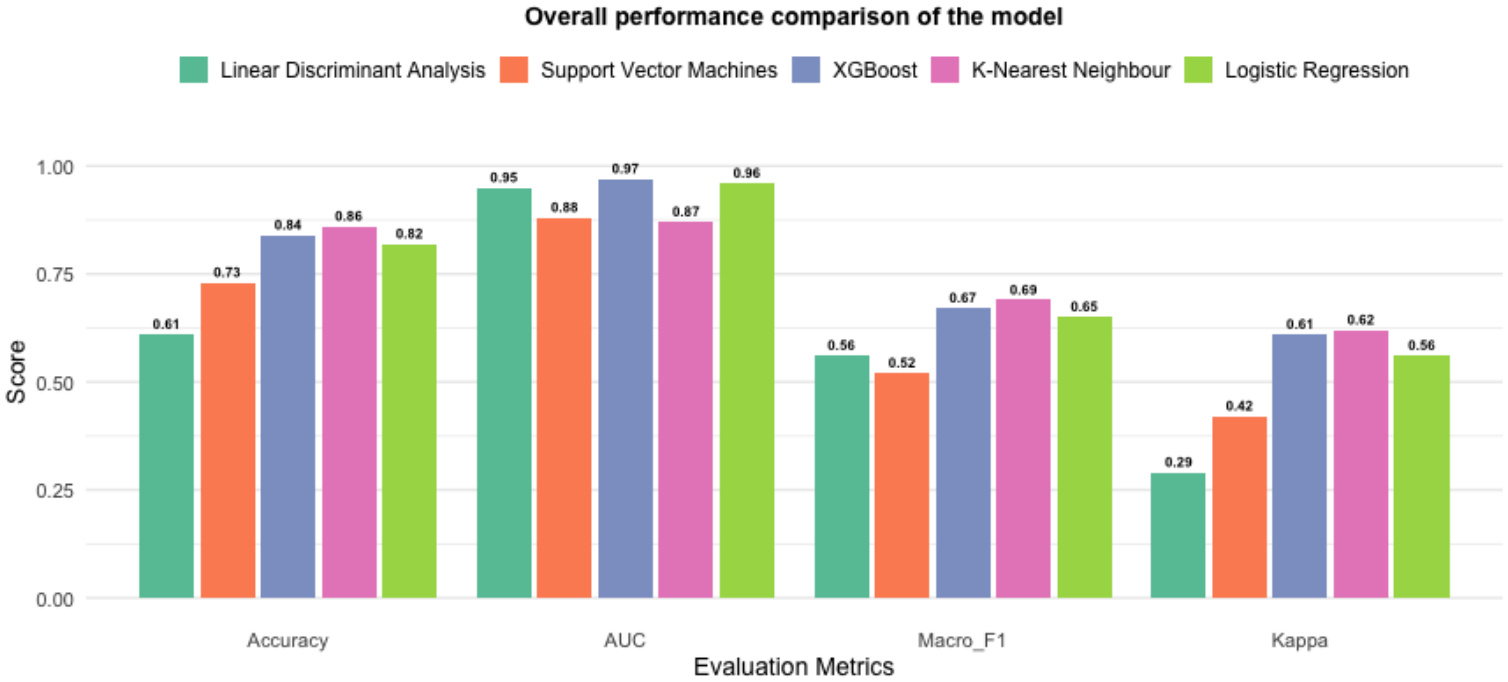
Key Results and Visuals



Classes Performance: All models perform good at almost all classes.

Weak Point: Poor is the most difficult class for all models, especially K-Nearest Neighbour.

Key Results and Visuals



Overall, XGBoost & K-Nearest Neighbour clearly outperform the others and are the top choices. Linear Discriminant Analysis performs weakest, but still workable.

Final Insights: Findings

Which factors significantly affect residential experience?

- **Multi-Faceted Nature:** The residential satisfaction is highly complex, driven by a combination of diverse factors (e.g., transport, safety, amenities) and differing demographic needs (backpackers vs. families).

How do different models perform in predicting residential experience?

- **Predicting User's Residential Experience:** The user's rating of residential experience is not only the user's subjective preference, but also a quantifiable and predictable signal.
- **Five-Class Classification:** For classes with more original data ("Excellent", "Good"), the five models differ little in accuracy. For the smaller classes ("Moderate", "Poor", "Very Poor"), even after SMOTE equalizes sample size, accuracies diverge substantially across models.

Final Insights: What Have We Learned

Incomplete data: Real-world data often contain missing values, noise, and inconsistent representations; high-quality preprocessing is critical to model performance.

Feature relationships: Not all features are relevant—removing irrelevant or redundant ones improves training efficiency and robustness.

Model selection: Models differ markedly in training time and performance; there is no "best" model, only the model that best fits the problem.

Thank you!

THE UNIVERSITY OF SYDNEY